AI251P Applied Computer Vision

Capstone Project – Phase 1 Report

Camera-Trap Image Classification

Project 5 M1–M3 28 October 2026

1 Team and project

Field	Answer
Team name	Project 5
Topic	Camera-trap image classification, Project 5
Application area	Wildlife monitoring and image triage
Code repository	https://github.com/salehdev3/Capstone_5
W&B project	https://wandb.ai/saleh-alanazi-ministry-of-interior/camera-trap-phase1
Dataset card	https://github.com/salehdev3/Capstone_5/blob/v1/DATASET_CARD.md

Team members

Name	ID	Workstream	Phase 1 contribution
Saud Abdullah Alfaiz	225899	Data and classification	Dataset acquisition, cleaning, split, audit and dataset card
Saleh Mohammed Alanazi	225486	CNN training	Scratch CNN, validation and test evaluation
Yousef Nafie Alanazi	225457	Transfer learning	Frozen and full ResNet-50 comparisons
Mohannad Mohammed Alqahtani	225461	Integration	Independent review, comparison and report integration

Problem and acceptance criteria

The system classifies full camera-trap frames as empty or animal, then classifies animal species on human-box crops. The output helps reviewers prioritize animal frames and provides a species baseline for the later pipeline. Recorded targets were animal recall ≥ 0.99, empty removal ≥ 0.20, species macro-F1 ≥ 0.65, and species ECE ≤ 0.10. Model selection used validation macro-F1 only.

2 Data foundations (M1)

2.1 Datasets

Dataset	Source	Use and annotations	Images
CCT20 subset	https://lila.science/datasets/caltech-camera-traps; source licence recorded as CDLA-Permissive	Camera-trap images and human boxes; train, validation and test	6,022
Team photographs	Riyadh and Tabuk; private team collection	14 Riyadh training frames and 5 Tabuk test frames; verified labels and boxes	19

The 6,022-image CCT20 subset was selected by the notebook configuration (seed 42); it is not the official CCT benchmark split. Team photos are a small convenience sample; original pigeon/chicken terms are retained in metadata and mapped to bird for modeling. Original images are kept private and omitted from the code repository.

2.2 Cleaning

Check	Method	Removed
Corrupt/unreadable	Download and decode verification	0
Exact duplicate IDs	ID/hash screening	0
Near-duplicate leakage	Conservative pHash screening across partitions	0
Human audit quarantine	One flagged training event, four frames	4
Remaining	6,041 initially kept; 6,037 after audit	6,037

2.3 Leakage-safe split

Camera groups keep correlated frames and bursts in one partition. A group-aware stratified split used seed 42; class proportions are approximate because cameras have different species mixes. Tabuk images remain in test and Riyadh images in train; CCT20 supplies validation and calibration. Automated ID, camera, event and pixel-hash cross-split checks passed.

Split	Groups	Images	Empty frames
Train	13	3,534	319
Validation selection	3	1,337	91
Validation calibration	2	527	13
Test	4	639	53

2.4 Label audit

Ten random training events were sampled, one frame each. One label error was found (10% observed rate); the recorded one-sided 95% upper bound is 39.4%. Event/camera correlation and

small sample size limit the estimate. A separate independent review of 20 images by Mohannad Mohammed Alqahtani recorded 20 correct frame labels, 15 correct boxes, five box checks not applicable, and no visible people. The no-box squirrel frame is triage-only. The independent review occurred after the frozen evaluation and does not change its reported metrics.

2.5 Data loader and augmentation

Images are RGB, prepared to a maximum side of 1,024 pixels, then resized to 224×224 and ImageNet-normalized for the model. Random flips and brightness/contrast changes apply to training only. Species classification uses human-box crops; full-frame triage also accepts animal images lacking boxes.

2.6 Dataset card summary

Class frequencies vary substantially by camera. Training has 764 raccoon and 727 opossum frame occurrences, but only one fox and four badger occurrences. Some rare classes are absent from validation groups. CCT20 is largely US wildlife imagery, whereas local photos are sparse and geographically narrow. The five Tabuk frames cannot establish Saudi deployment performance. Grayscale appearance is only an infrared proxy; no GPS is published. The full card is linked in Section 1.

3 Classification (M2–M3)

Tasks: full-frame empty/animal classification, followed by species classification on human-box crops. All runs use 224×224 input. Frozen and full transfer runs use pretrained ResNet-50; scratch uses a three-block CNN. Validation macro-F1 selects the frozen regime for each task.

3.1 Runs compared

Task	Regime	Epochs	Val. macro-F1	Test acc.	Test macro-F1
Triage	Scratch CNN	6	.6011	.9171	.4784
Triage	Frozen ResNet-50	12	.6617	.9327	.6515
Triage	Full ResNet-50	12	.6404	.9092	.5943
Species	Scratch CNN	12	.1557	.3504	.1780
Species	Frozen ResNet-50	12	.3641	.7421	.4820
Species	Full ResNet-50	12	.3398	.7638	.4546

W&B run links are recorded in `results/wandb_links.csv`. Test accuracy camera-bootstrap intervals for selected triage and species are [.897,.967] and [.642,.876], respectively; the intervals are descriptive with only four held-out cameras. Selected test ECE is .0460 for triage and .0568 for species. Top-3 species accuracy is .9055. Full fine-tuning has slightly higher species accuracy, but frozen was selected on validation macro-F1 and has higher test macro-F1.

3.2 Learning curves

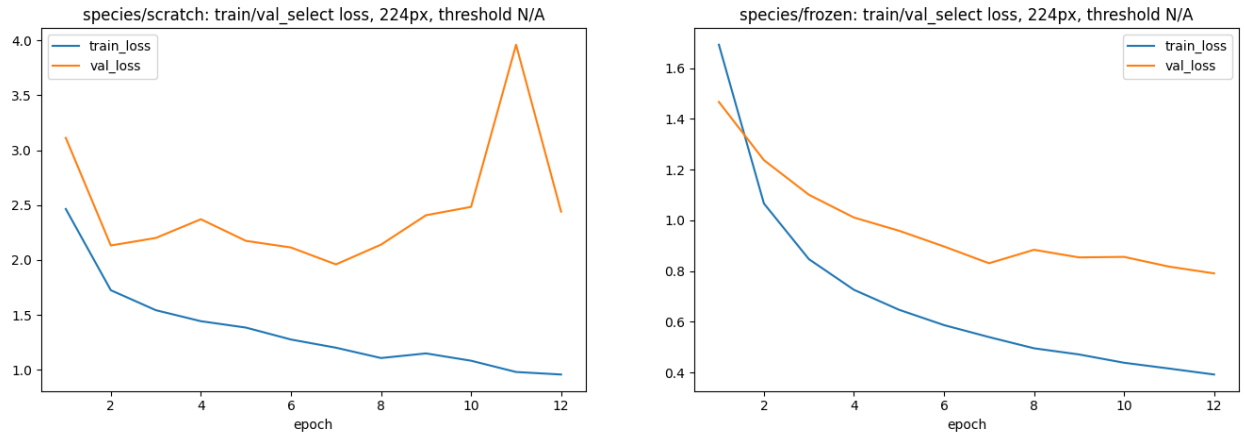


Figure 1: Species training and validation-selection loss, 224×224 input, argmax decision; scratch CNN (left), frozen ResNet-50 (right).

The scratch species model's training accuracy increased while held-out macro-F1 remained low. Frozen features improved validation macro-F1 from .1557 to .3641. Rare classes and camera-domain shift remain limitations.

3.3 Per-class results (selected models, test set)

Class	Precision	Recall	F1	Support
Empty (triage)	.917	.208	.338	53
Animal (triage)	.933	.998	.965	586
Bird	.900	1.000	.947	9
Bobcat	.629	.684	.655	57
Cat	.632	.250	.358	48
Coyote	.956	.763	.849	169
Dog	.676	.862	.758	29
Opossum	.656	.966	.781	87
Raccoon	.366	.536	.435	28
Squirrel	.860	.925	.891	53
Badger / deer / fox	.000	.000	.000	1 / 3 / 1
Species macro average	.490	.498	.482	508

Additional test classes and their support appear in `results/species/frozen/test_metrics.json`. The selected triage operating threshold is .3416: 586/586 animals retained, 6/53 empty frames removed (11.3%). It meets the 99% animal-recall target and misses 20% empty-removal. The selected species model meets ECE $\leq$.10 and misses macro-F1 $\geq$.65.

3.4 Confusion matrix and calibration

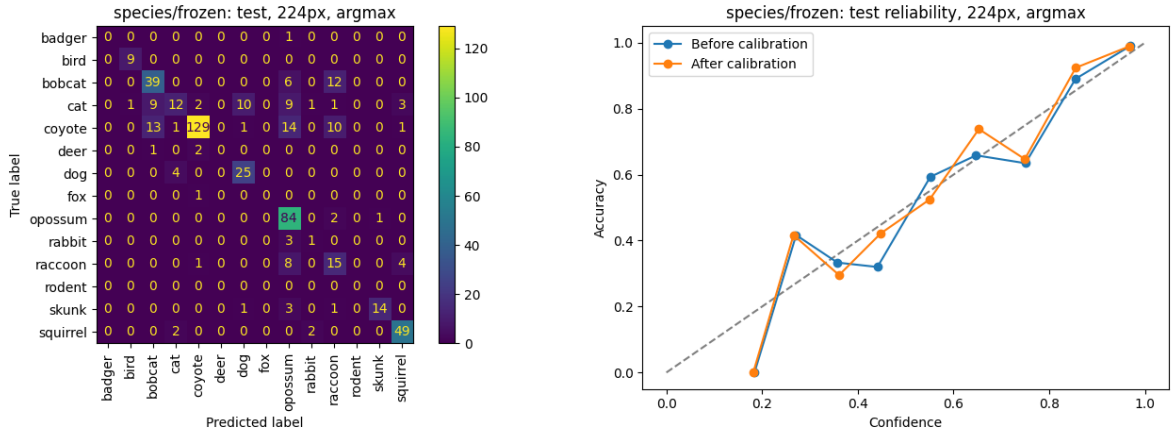


Figure 2: Selected frozen species model on test crops, 224×224 input, argmax. Confusion matrix (left) and reliability diagram before/after validation-calibration temperature scaling (right).

Species test ECE is .0568; the uncalibrated ECE is .0454, so calibration does not improve ECE on this test split. The model confuses bobcat with raccoon and some cats with other mammals. These are observations from a small subset of high-confidence errors, not a complete causal diagnosis.

3.5 Error analysis

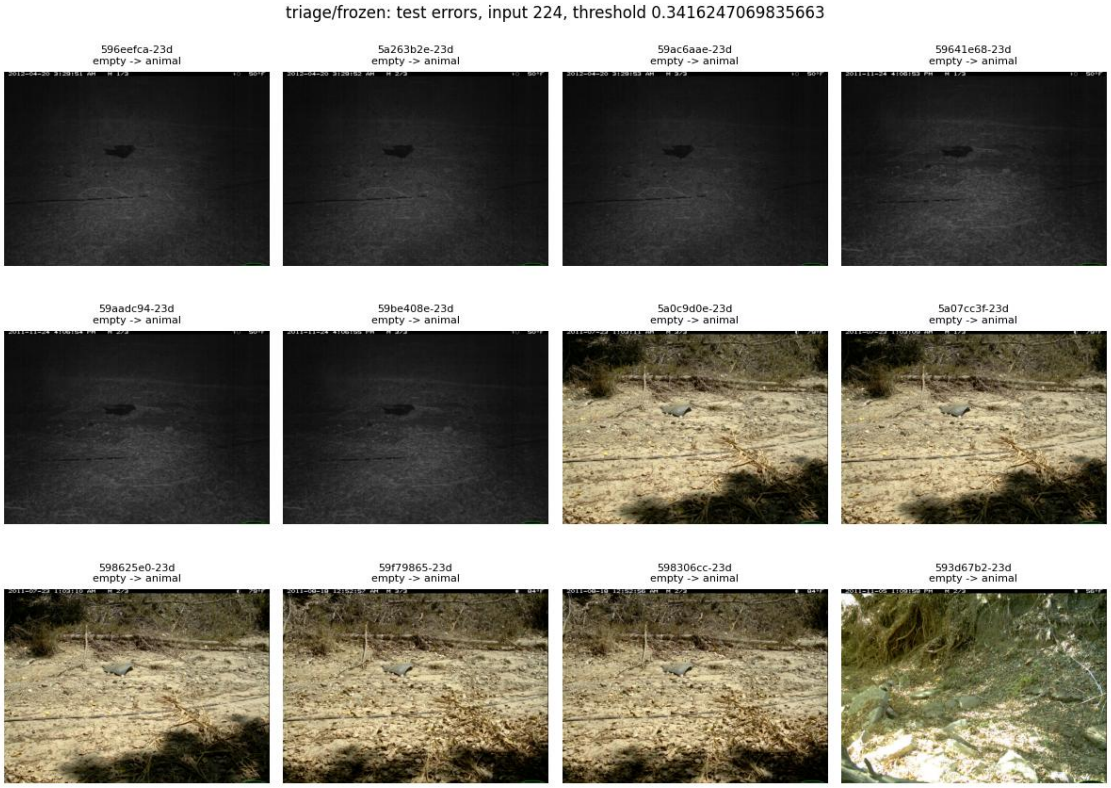


Figure 3: Selected triage test error gallery, 224×224 input, animal threshold .3416. Recorded empty frames predicted animal; original labels and full-resolution images require human adjudication.

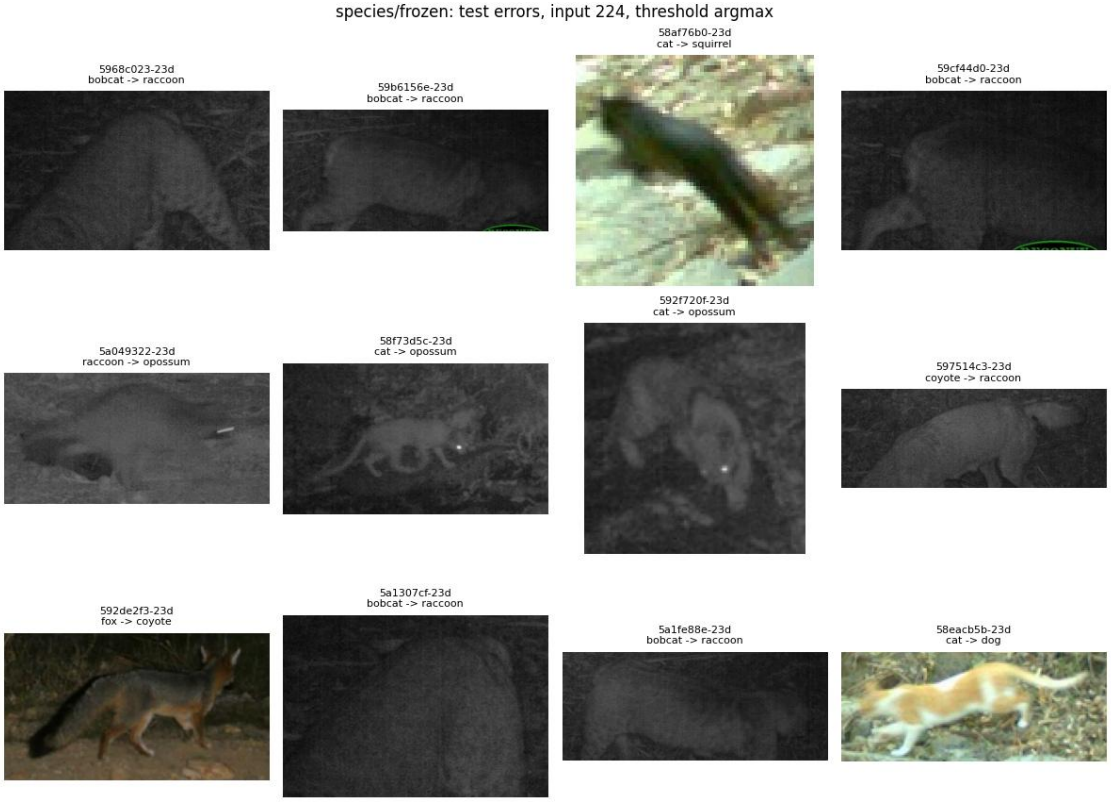


Figure 4: Selected species test error gallery on human-box crops, 224×224 input, argmax. Titles show recorded label and prediction; sample IDs link to private originals.

Several daylight triage images recorded as empty appear to contain birds, while dark silhouettes need full-resolution inspection. In the species gallery, five displayed bobcat crops were predicted raccoon; one fox crop was predicted coyote. A crop recorded as cat (ID prefix `58af76b0`) looks bird-like and needs source-box/label review. We retain original locked-test results and do not silently relabel the score.

3.6 Data-improvement plan

Finding	Planned action	Expected effect
Sparse rare species	Obtain independently grouped, verified examples	Better rare-class recall and more informative validation
Possible test-label errors	Human-adjudicate originals and report corrected-label sensitivity separately	Separate annotation error from model error
Empty-frame removal 11.3%	Add diverse empty training frames; tune only on validation calibration	Improve removal while preserving recall

The separate five-frame Tabuk subset is reported descriptively: triage retained both animal frames but removed none of three empty frames at threshold .3416; species got 4/5 known crops correct. It is too small to estimate deployment accuracy.

4 Reproducibility

Item	Evidence
Repository	https://github.com/salehdev3/Capstone_5, branch `v1`; evaluated dataset hash `d3bfb6ba0c9d31412f9d603a6f63ae7c4890f458c5d2aa2a87d7653bb882bb53`
Configuration	`handoff/data_config.json`, role `runs/member2/config.json` and `runs/member3/config.json`
Random seed	42
Environment	Pinned role requirements and runtime records under `handoff/`
Reproduction	Execute `notebooks/Phase1_Final_Presentation.ipynb` in section order on Colab GPU; use the same released manifest and configuration
Data source	CCT20 from LILA and private team photos; images not committed
Independent review	Mohannad Mohammed Alqahtani, 20-image review; fresh-process inference checks are recorded in `review/`

The exported evidence includes each run's summary, history, policy and test metrics plus W&B IDs. The team should confirm instructor access to the private repository, W&B project and dataset-card link. A named clean full-notebook replay is still pending; fresh-process inference alone does not establish full training reproducibility.

5 Submission checklist

- ✓ Corrupt/duplicate checks, grouped split, label audit and Dataset Card recorded.
- ✓ Scratch and transfer classifiers compared; validation selected models; per-class metrics and ECE reported.
- ✓ Figures state split, 224×224 input and threshold/argmax.
- ☐ Confirm instructor access to all three links and full clean replay evidence.
- ☐ Human-adjudicate suspicious test labels and document any corrections separately.

6 Self-assessment

Criterion	Points	Evidence	Known gap
M1 data foundations	12	Cleaning, grouped split, audit, card	Small audit sample; independent review post-run
M2–M3 classification	20	Six runs, test metrics, galleries	Two preset performance targets missed
Reproducibility	8	Config, requirements, hashes, W&B	Full clean replay and link access pending
Total	40	Self-score deferred until final checks	–

7 Obstacles and how they were addressed

Obstacle	Module	Impact	Action	Status
Colab runtime and W&B upload stalls	M2–M3	Interrupted evaluation workflow	Save local test metrics before bounded upload	Resolved in code
Rare classes and camera-domain shift	M1–M3	Low species macro-F1	Grouped split, report supports and improvement plan	Open
Uncertain source annotations	M1–M3	Some apparent errors may be label errors	Independent audit and proposed test-gallery adjudication	Open